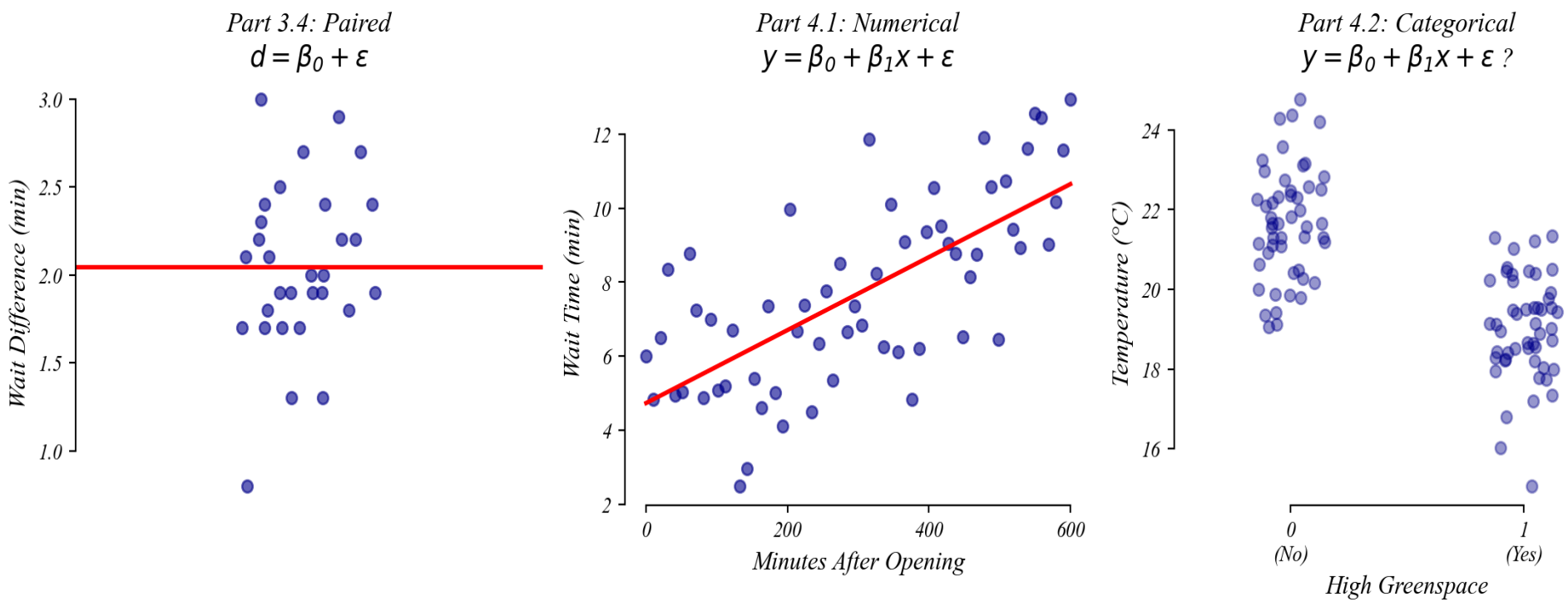
ECON 0150 | Economic Data Analysis

The economist's data analysis workflow.

Part 4.2 | Categorical Predictors

GLM: The story so far

We've built two models. Now we need a third.

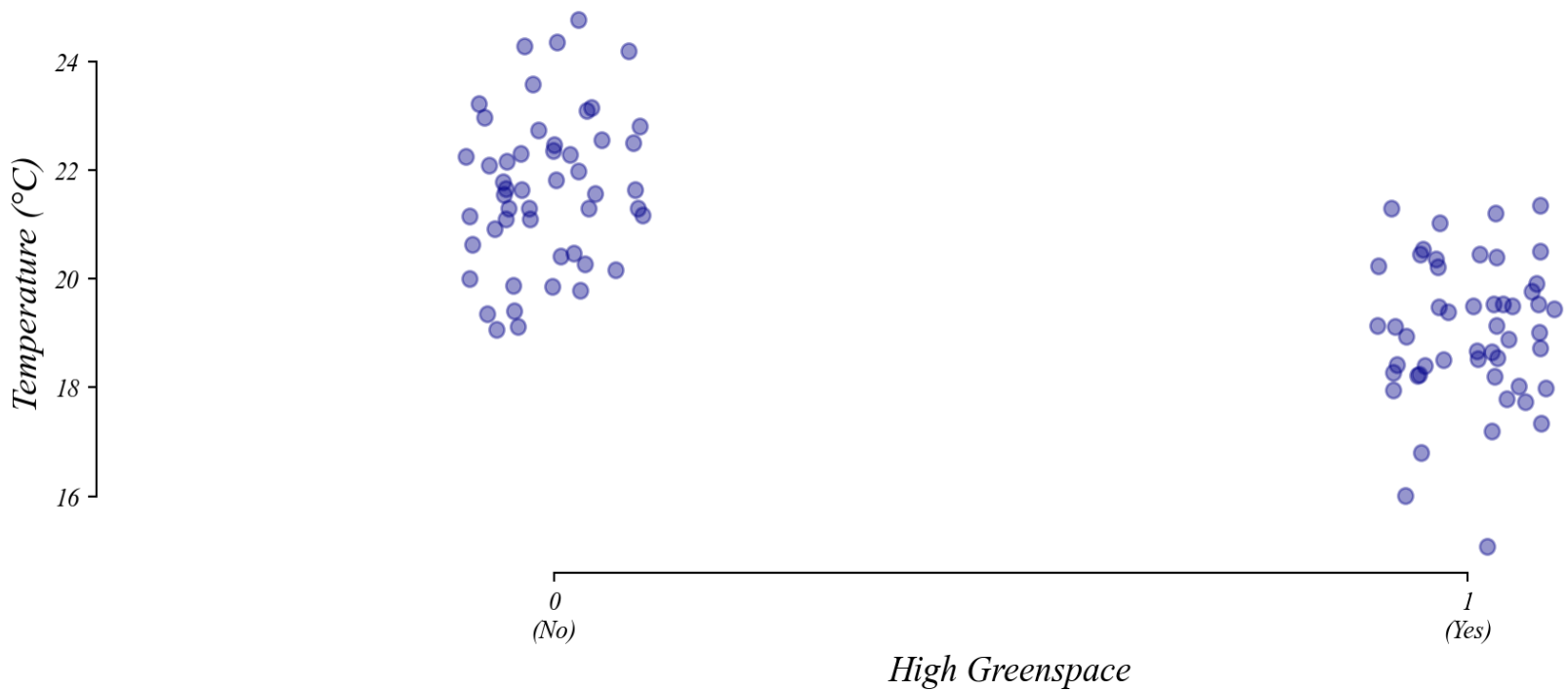


Q. *Do neighborhoods with greenspace have lower temperatures?*

We need a new approach, but we already have the tools.

GLM: City Greenspace and Temperature

Q. Is temperature lower in neighborhoods with more green space?



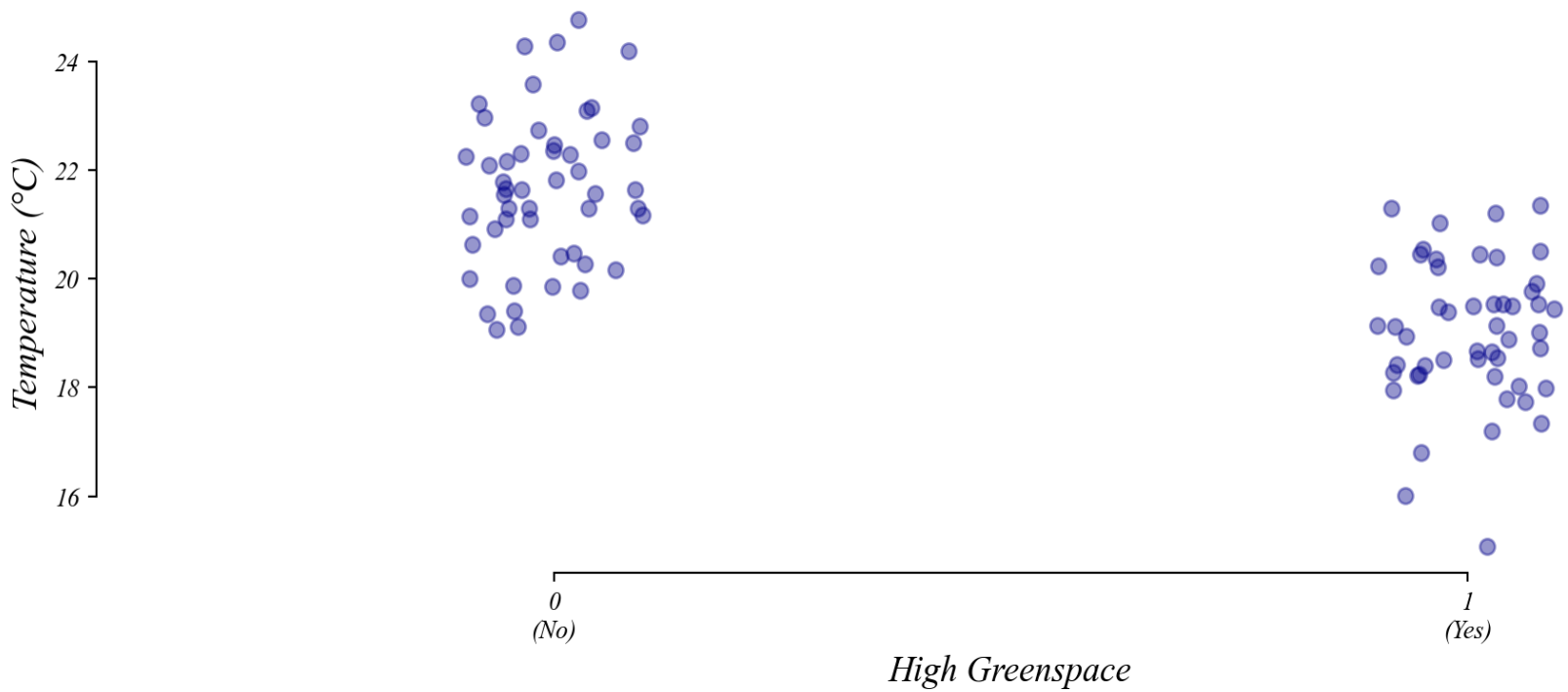
*What is the variable type of **High Greenspace**?*

***Q.** Does temperature change as we increase by one on the horizontal axis?*

$$Temperature = \beta_0 + \beta_1 \cdot HighGreen + \varepsilon$$

GLM: City Greenspace and Temperature

Q. Is temperature lower in neighborhoods with more green space?

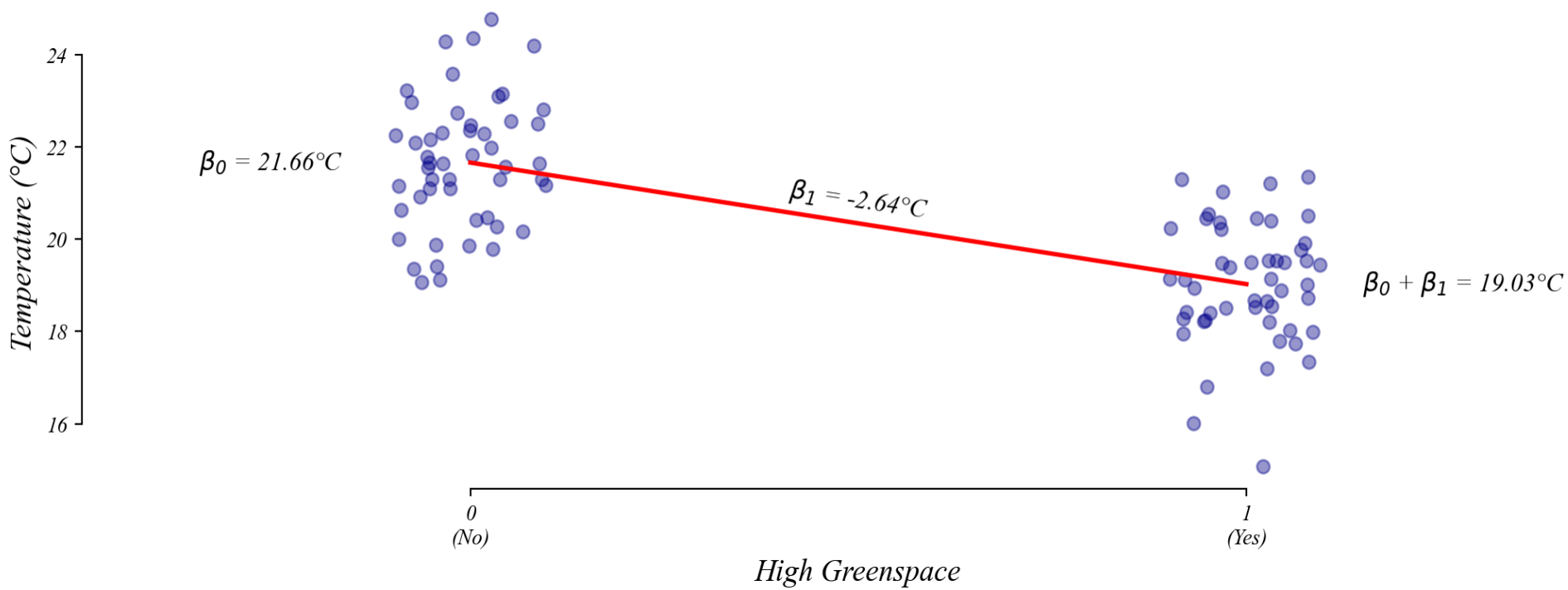


1. *The GLM framework doesn't care whether* x *is continuous or binary. We code the group as 0 and 1, and fit the same model.*

2. *Like with numerical predictors, the GLM tests whether* β_1 *is different from zero.*

GLM: City Greenspace and Temperature

Q. Does temperature change as we move out on the horizontal axis?



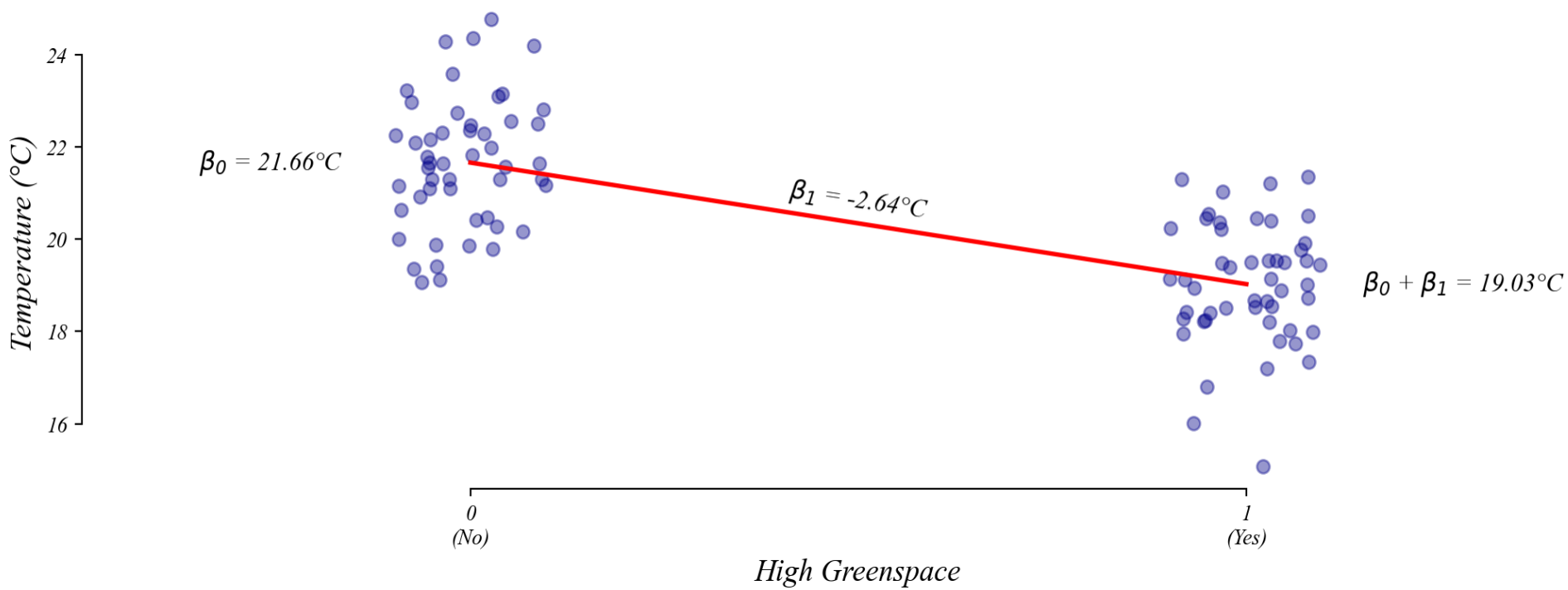
$$Temperature = \beta_0 + \beta_1 \cdot HighGreen + \varepsilon$$

How would we interpret β_0 here?

> *β_0 is the mean temperature in ($x = 0$) low green space cities (22.03°C)*

GLM: City Greenspace and Temperature

Q. Does temperature change as we move out on the horizontal axis?



$$Temperature = \beta_0 + \beta_1 \cdot HighGreen + \varepsilon$$

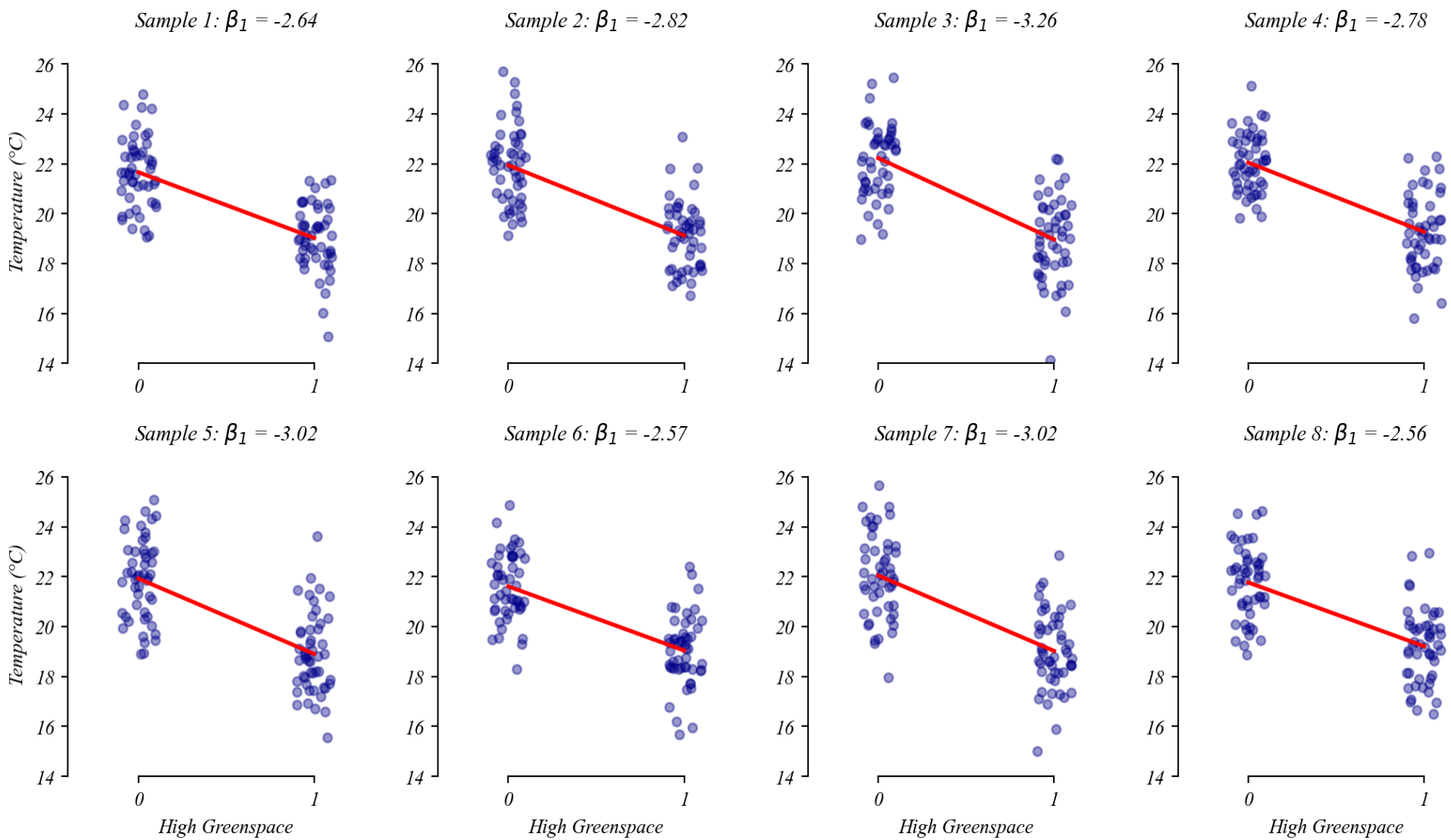
How would we interpret β_1 here?

> *Cities with Green Space (x=1) have a temperature that is lower by* β_1

> *ie. a one unit increase in x changes temperature by β_1*

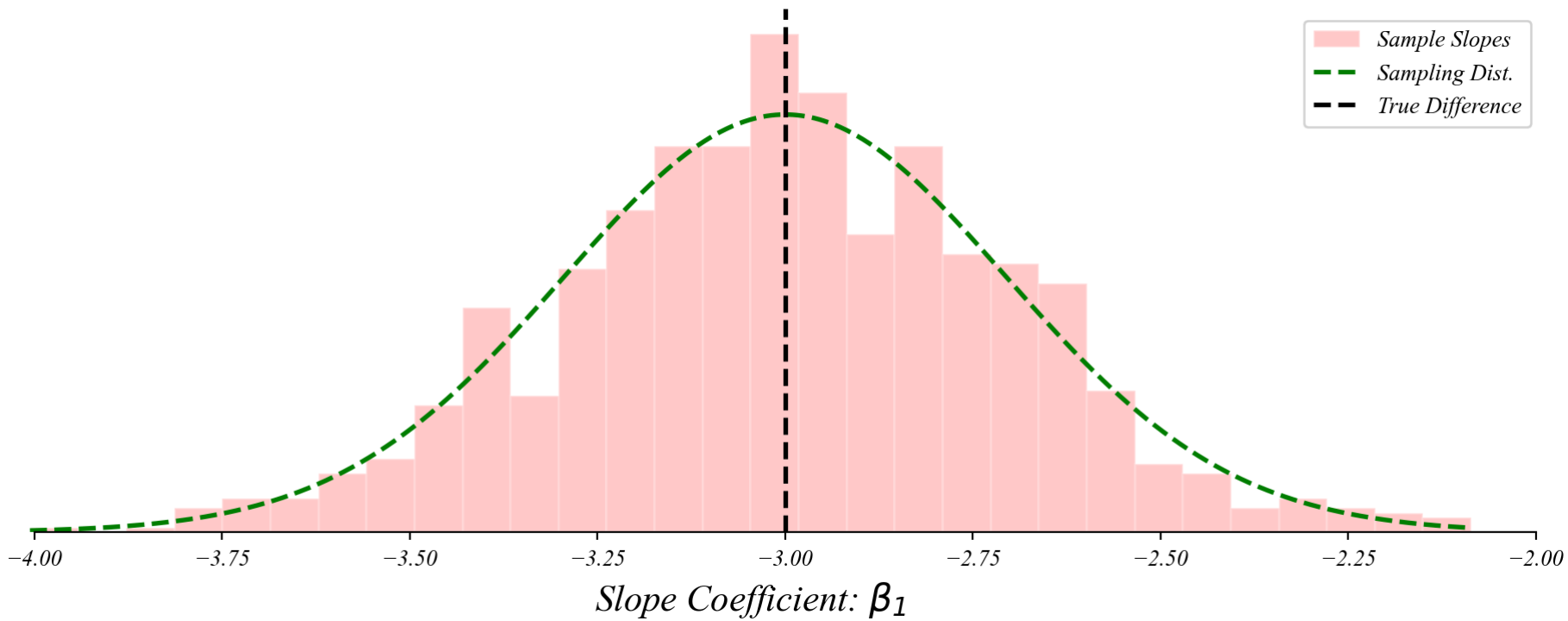
Categorical GLM: sampling error

Like before, if we take many samples, we get slightly different slopes and slightly different fits.



Categorical GLM: sampling distribution of slopes

The slope coefficient follows a normal distribution centered on the population difference.

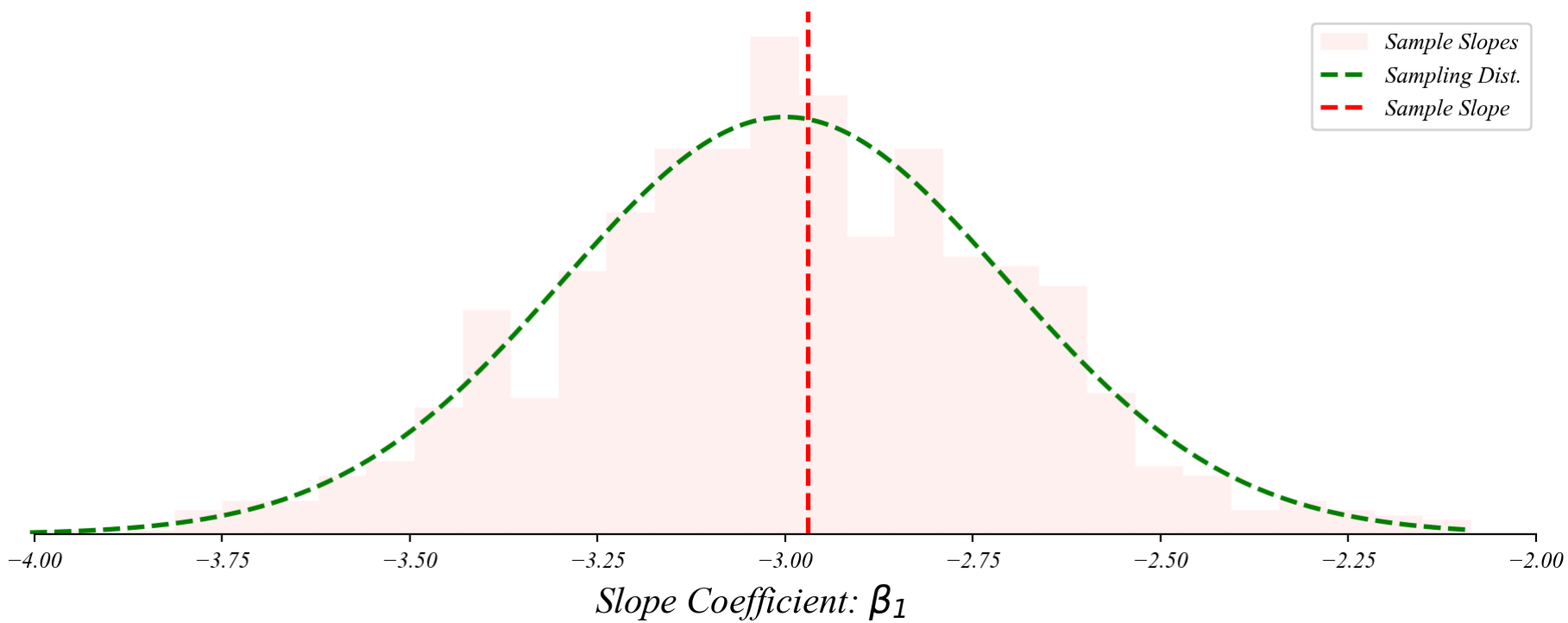


> *the slopes follow a normal distribution around the population difference!*

> *this lets us perform a t-test on the slope!*

Categorical GLM: sampling distribution of slopes

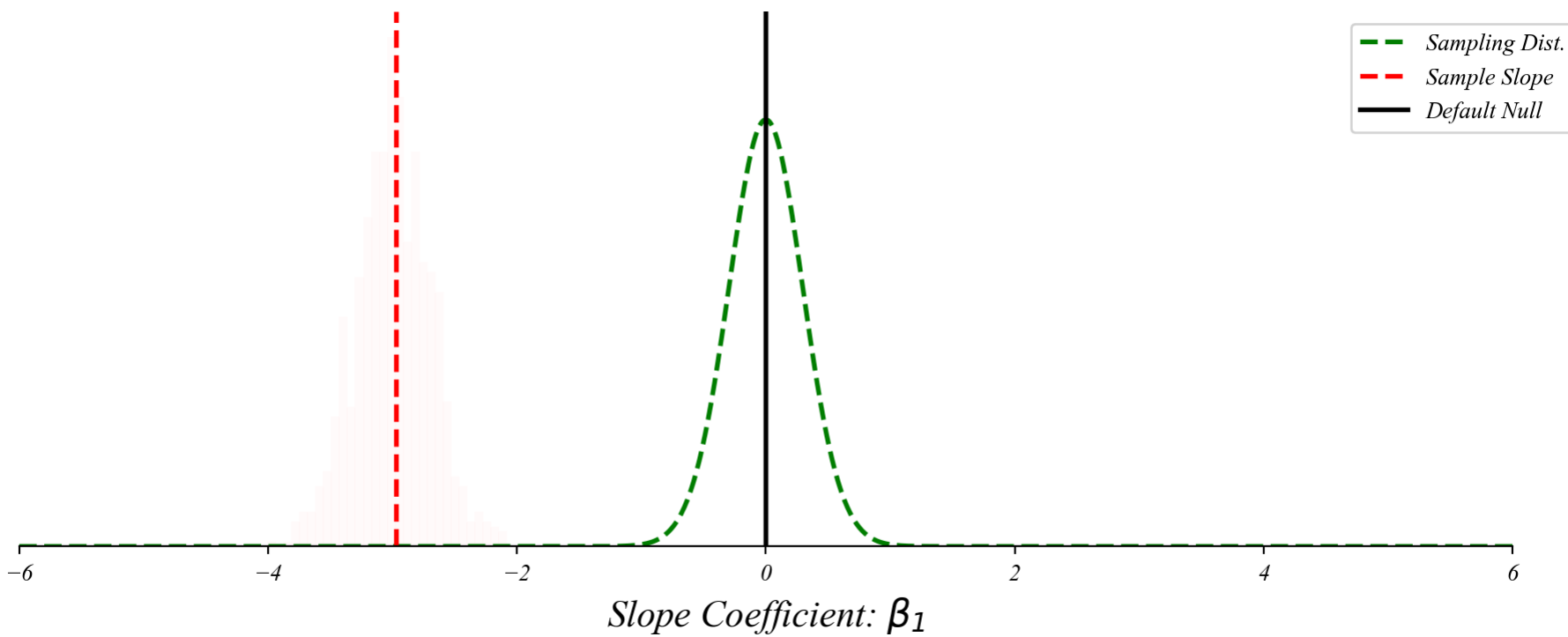
The slope coefficient follows a normal distribution centered on the population difference.



> *we don't know the entire distribution, just our sample slope*

Categorical GLM: sampling distribution of slopes

The slope coefficient follows a normal distribution centered on the population difference.

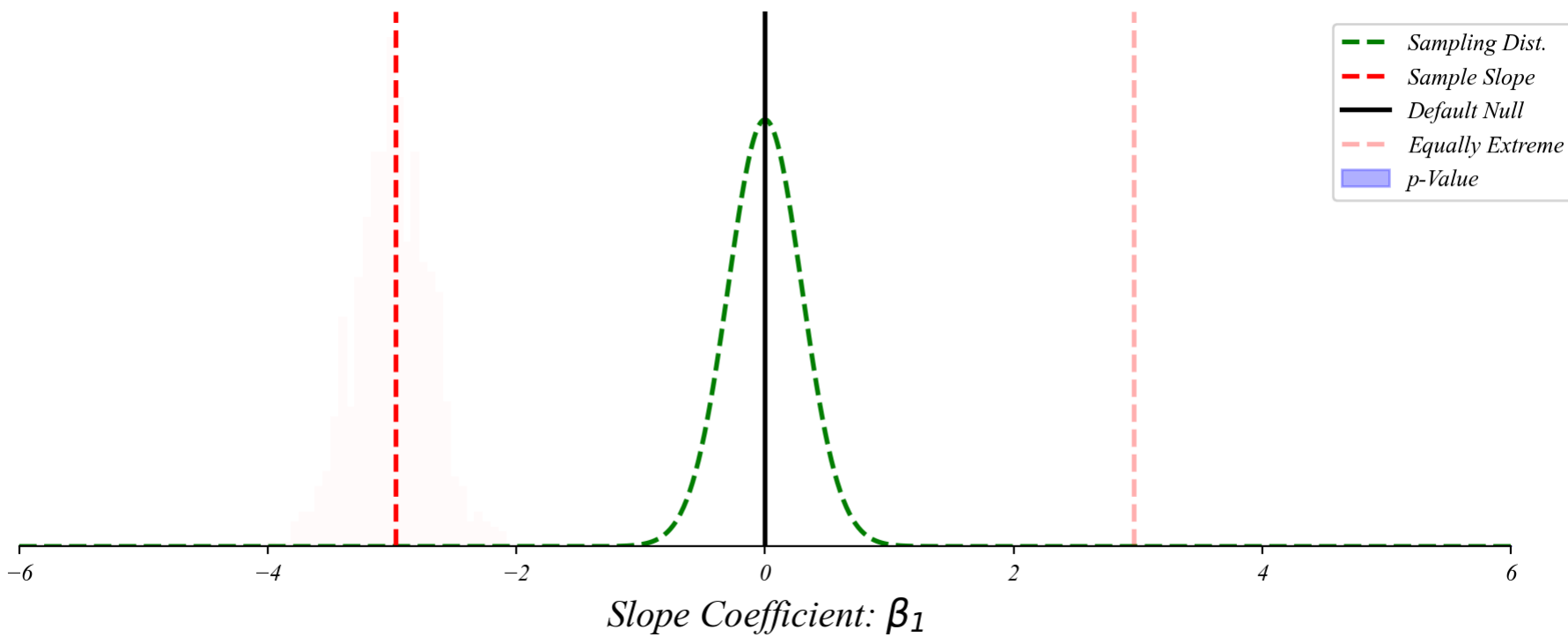


> *center the distribution on our null*

> *check the distance from the sample*

Categorical GLM: sampling distribution of slopes

The slope coefficient follows a normal distribution centered on the population difference.



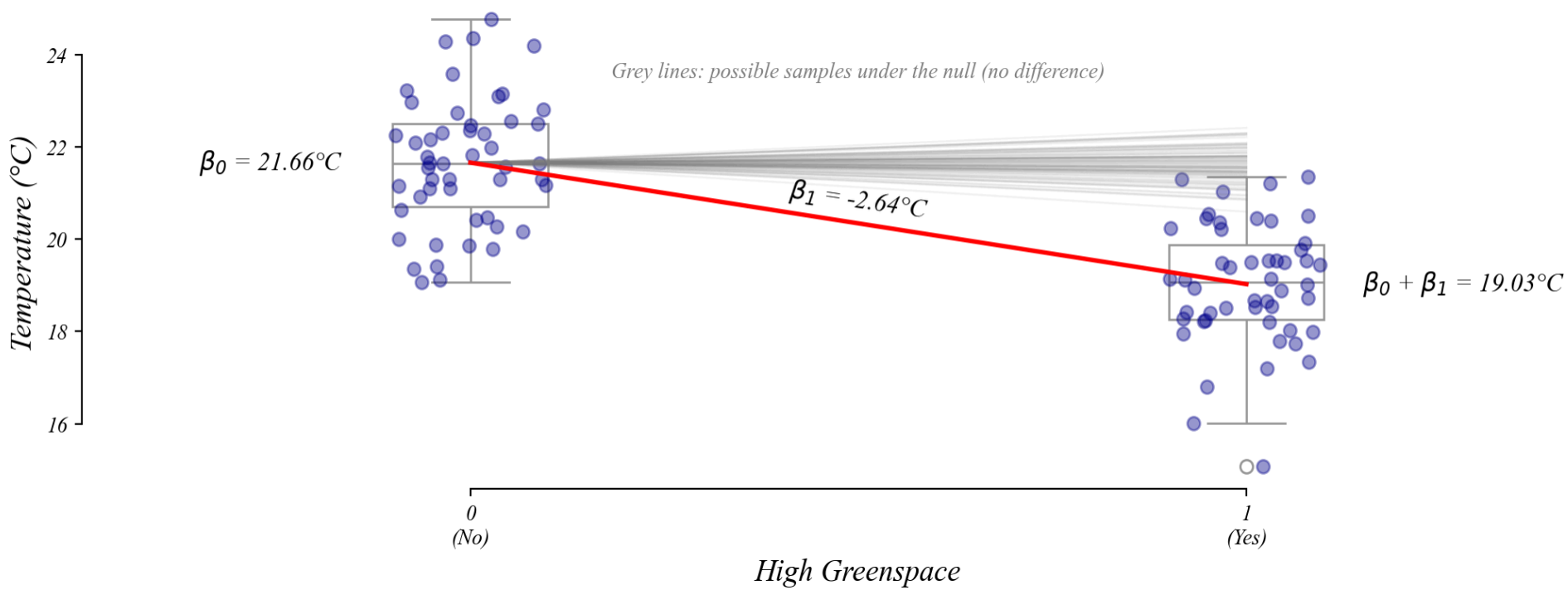
> *p-value: the 'surprisingness' of our sample if* $\beta_1 = 0$

> *the probability of seeing our sample by chance if there is no difference*

> *a small p-value is evidence against the null hypothesis* ($\beta_1 = 0$)

Categorical GLM: null slopes

Many possible slopes we might observe by chance if the null ($\beta_1 = 0$) were true.



> *how likely does it look like this slope was drawn from the null slopes?*

> *p-value: the probability of a slope as extreme as ours under the null ($\beta_1 = 0$)*

This is a two-sample t-test

This is a simpler view of a t-test.

1. *Estimating differences between group means is exactly a* ***two-sample t-test****.*

2. *GLM is a general framework that can handle in one modeling approach:*

- *Numerical predictors*
- *Categorical predictors*
- *Control variables (later)*

3. *The two-sample t-test is a special case of the GLM.*

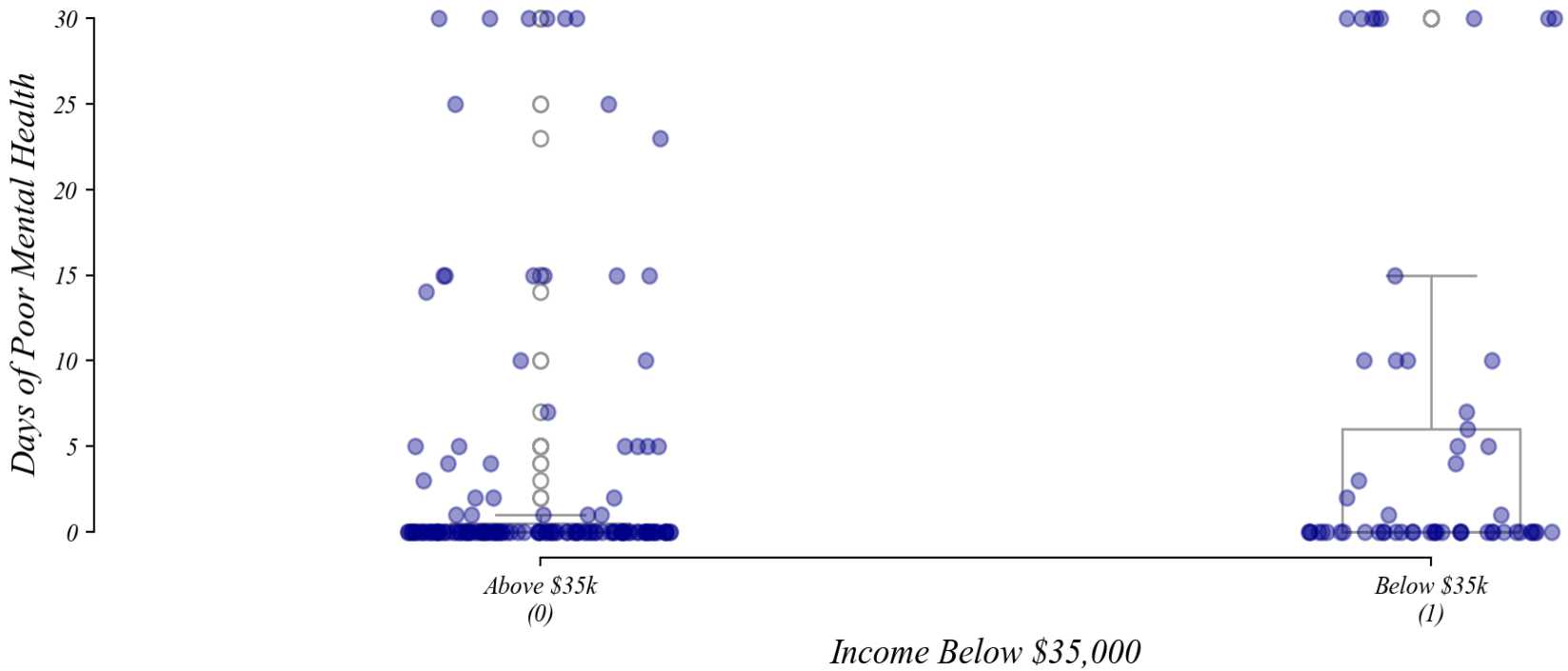
Exercise: Income and Mental Health

Do individuals earning under $35,000 report more poor mental health days?

Step 1: Summarize the data

```
# Load BRFSS data
data = pd.read_csv('data/BRFSS_cleaned.csv')
data['low_income'] = (data['INCOME2'] <= 5).astype(int)

# Visualize Binary Predictor
sns.boxplot(data=data, x='low_income', y='MENTHLTH', color='white', width=0.2)
plt.xticks([0,1], labels=['Above $35k', 'Below $35k']  # INCOME2 <= 5)
```


Days of Poor Mental Health
30
25
20
15
10
5
0
Above $35k
(0)
Below $35k
(1)
Income Below $35,000

Exercise: Income and Mental Health

Do individuals earning under $35,000 report more poor mental health days?

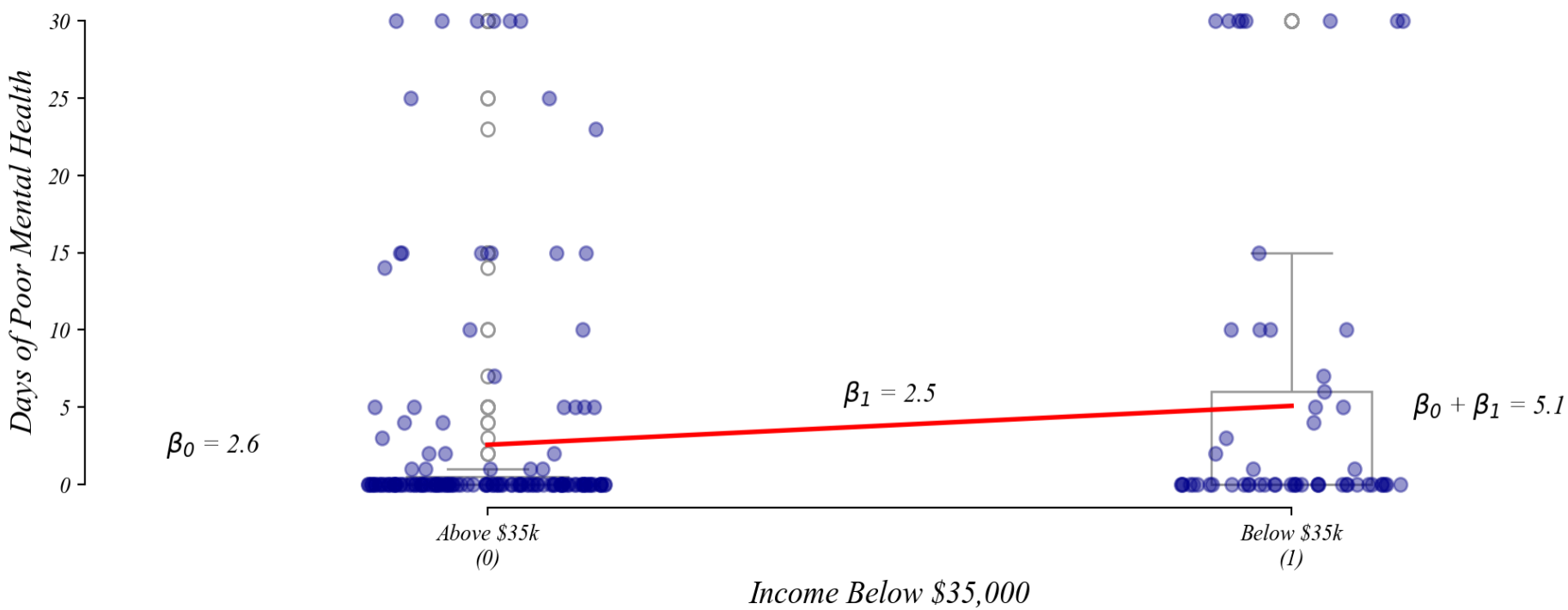
Step 2: Build a model

$$MentalHealthDays = \beta_0 + \beta_1 \cdot LowIncome + \varepsilon$$

Exercise: Income and Mental Health

Do individuals earning under $35,000 report more poor mental health days?

Step 3: Estimate the model




```
# Model: y = b + mx
model = smf.ols('MENTHLTH ~ low_income', data).fit() # Intercept is included by default
print(model.summary().tables[1])
```

- β_0 = *Mean poor mental health days for those earning above $35k*
- β_1 = *Additional poor mental health days for those earning below $35k*

Exercise: Income and Mental Health

Do individuals earning under $35,000 report more poor mental health days?

Step 4: Check the residuals (next time)

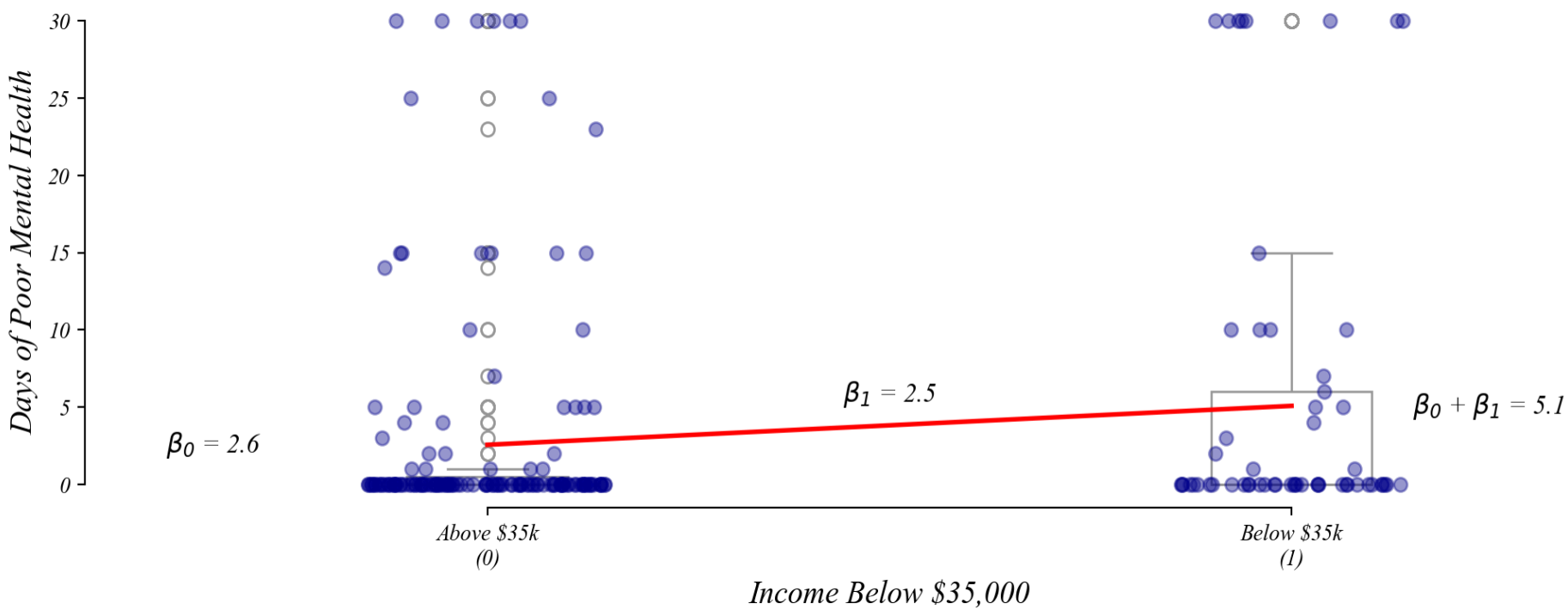
```
sns.scatterplot(x=model.predict(), y=model.resid, alpha=0.5)
plt.axhline(y=0, color='red', linestyle='-')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
```

> *what do you notice about this residual plot?*

Exercise: Income and Mental Health

Do individuals earning under $35,000 report more poor mental health days?

Step 5: Interpret and communicate the findings



> *A significant positive β_1 suggests those earning under $35k report more poor mental health days per month*

> *but does low income cause poor mental health, or are other factors at play?*

The General Linear Model *(so far)*

Extending the GLM framework

- *Part 3 | Intercept-Only Model* | $y = \beta_0 + \epsilon$
- *Part 4.1 | Numerical Predictor* | $y = \beta_0 + \beta_1 x + \epsilon$
- *Part 4.2 | Categorical Predictor* | $y = \beta_0 + \beta_1 x + \epsilon$
- *Part 4.3 | Model Diagnostics*
- *Part 4.4 | Causality*

Looking Forward

Extending the GLM framework

Part 5 | Control Variables

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \varepsilon$$

- *Part 5.1 | Numerical Controls*
- *Part 5.2 | Categorical Controls*
- *Part 5.3 | Interactions*
- *Part 5.4 | Model Selection*

> *all built on the same statistical foundation*